

AVERAGES

A list of numbers or numerical data can be described by various statistical measures. One of the most common of these measures is the Average or Arithmetic Mean, which locates a type of center for the data.

Arithmetic Mean may be either simple arithmetic mean or weighted arithmetic mean. The weighted arithmetic mean is also called as compound average or combined average.

The arithmetic mean of n numbers of data is defined as the sum of n numbers divided by

$$\text{Arithmetic mean} = \bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{\sum_{i=1}^n X_i}{n}$$

Weighted Arithmetic Mean. The term 'weight' stands for the relative importance of the different terms.

Weighted Arithmetic Mean = $\left\{ \frac{\sum_{i=1}^n w_i * \sum_{i=1}^n x_i}{\sum_{i=1}^n w_i} \right\}$ Where W is weight and X is a variable

Examples

1) The average age of a class of 26 students is 14 years, A new student joined the class and the average of the class increased by 4 months. The age of the new student is

- A) 18 years B) 19 years C) 21 years
D) 23 years E) 22 years

Let the age of the new student be 'A' years

$$\therefore (26 \times 14) + A = 27 (14 + 1/3), A = 387 - 364 = 23 \text{ (D)}$$

Short – Cut Method

$$\text{Average increase} = 1/3 \text{ years}$$

$$\text{Net increase} = 27 \times 1/3 = 9$$

$$\text{Age of the new student is } 14 + 9 = 23 \text{ years. (D)}$$

2) The girls' average score is 40. The boys' average score is 52. If there are thrice as many girls as boys, what is the overall average?

- A) 46 B) 43 C) 42 D) 48 E) 44

$$\text{Weighted Average} = \frac{3(40) + 1(52)}{4} = 43 \text{ (B)}$$

3) The average of 6 numbers is 8.5. When one number is discarded, the average of the remaining numbers becomes 7.2. What is discarded number?

- A) 7.8 B) 9.8 C) 10 D) 12.4 E) 15

$$\text{Discarded number} = 6(8.5) - 5(7.2) = 51 - 36 = 15 \text{ (E)}$$

EXCERSICE

1. The average of a batsman after 30 innings was 70 runs per innings. If after the 31st innings, his average increased by 2 runs, then what was his score in the 31st innings?

- a) 136 b) 132 c) 127 d) 140

2. The average weight of A, B and C is 45 kg. If the average weight of A and B be 40 kg and that of B and C be 43 kg, then the weight of B is:

- a) 17 kg b) 20 kg c) 26 kg
d) 31 kg e) None of these

3. The average age of 30 students is 9 years. If the age of their teacher is included, it becomes 10 years. The age of the teacher is

- a) 27 years b) 31 years c) 35 years
d) 40 years e) None of these

4. Average temperature for a week is noted as 32°C but later it was realized that one day's reading was taken as 30°C instead of 16°C . What is the correct average temperature for the week?

- a) 14°C b) 16°C c) 30°C d) 34°C

5. The average of 7 consecutive numbers is 20. The largest of these numbers is:

- a) 24 b) 23 c) 22
d) 20 e) None of these

6. The average of four consecutive even numbers is 27. The largest of these numbers is

- a) 36 b) 32 c) 30
d) 28 e) None of these

7. The average of 5 positive integers is 436. The average of the first two numbers is 344 and the average of the last two numbers is 554. What is the third number?

- a) 428 b) 540 c) 384
d) 632 e) None of these

8. The average of five results is 46 and that of the first four is 45. Find the fifth result?

- a) 48 b) 50 c) 52 d) 46 e) None of these

9. The average weight of 8 persons is increased by 2.5 kg when one of them whose weight is 56 kg is replaced by a new man. Find the weight of the new man?

- a) 70 b) 60 c) 66 d) 76 e) None of these

10. A cricketer has completed 10 innings and his average is 21.5 runs. How many runs must he make in his next innings so as to raise his average to 24?

- a) 29.5 b) 30 c) 25 d) 49 e) None of these

11. What is the average of first five multiples of 5?

- a) 20 b) 25 c) 15 d) 10

12. If a student secured 73% in English, 92% in Mathematics, 66% in Hindi, 54% in sociology and 85% in Economics, what will be his overall average?

- a) 73% b) 75% c) 74% d) 77%

13. The population of six villages is 803, 900, 1100, 1023, 945 and 980. What is the population of the seventh village, if the average population of seven villages is 1000?

- a) 1249 b) 1429 c) 1428 d) 1349

14. The average weight of students in a class of 35 is 50Kg. However if the weight of the teacher be included, the average will be increased by 600gm. Find the weight of the teacher.

- a) 71.2 b) 70 c) 71 d) 71.6

15. The average salary of 25 workers in a factory is Rs.1900 per month. If the manager's salary is added, the average salary becomes 2100 per month. The manager's average salary (in Rs) is

- a) 7200 b) 7100 c) 7300 d) 7000

ASSIGNMENT

16. The average of 180 third grade student is 12 years. The average of 40 section A students is 10 years, 40 section B is 14 years and 50 section C students is 12 years. What is the average age of section D students?

- a) 11 b) 12 c) 13 d) 14

17. The average of 11 numbers is 109. If the average of first six numbers is 105 and that of last six numbers is 114. What is the sixth number?
a) 116 b) 125 c) 115 d) 110
18. The average weight of a group of 50 girls was calculated as 58kg. It was later discovered that the weight of one of the girl was read as 45 kg. Whereas her actual weight was 65kg. What is the average of the group?
a) 58.4 b) 60 c) 55.5 d) 57
19. A batsman makes a score of 67 runs in the 17th innings and thus increases his average by 3. Find his average after 17th innings.
a) 20 b) 18 c) 19 d) 21
20. Distance between two stations A&B is 778 km. A train covers the journey from A to B at 80 km/hr. and return back to A with a uniform speed of 56km/hr. Find the average speed of train during the journey.
a) 64.8 km/hr. b) 65.8 km/hr.
c) 65.2 km/hr. d) 64.2 km/hr.
21. A Library has an average of 400 visitors on Sundays & 190 on other days. The average number of visitors per day in a month of 30 days beginning with a Sunday is
a) 235 b) 205 c) 215 d) 225
22. The batting average for 30 innings of a cricket player is 50 runs. His highest score exceeds his lowest score by 152 runs. If these two innings are excluded, the average of the remaining 28 innings is 48 runs. The highest score of the player is
a) 144 b) 154 c) 152 d) 158
23. The mean of 50 observations was 36. It was found later that an observation 48 was wrongly taken as 23. The corrected new mean is:
a) 30.5 b) 36.5 c) 37
d) 39.5 e) None of these.
24. The average age of a family of 6 members is 22 years. If the age of the youngest member be 7 years, then what was the average age of the family at the birth of the youngest member?
a) 16 b) 18 c) 20 d) 22
25. The average age of 30 boys of a class is equal to 14 years. When the age of the class teacher is included the average becomes 15 years. Find the age of the class teacher?
a) 40 b) 43 c) 45
d) 47 e) None of these

ALLIGATION

Mixtures:

In dilution or mixture problems, we come across the following type of questions.

To determine the characteristic of the resulting mixture when substance with different characteristics are combined.

To determine how the combined substances with different characteristics will produce a desired mixture.

Example.

1. How many liters of a solution, that is 10% alcohol by volume, must be added to 2 liters of a solution that is 50% alcohol by volume to make a solution is 15% alcohol by volume?

- a) 14 liters b) 16 liters c) 40 liters
d) 20 liters e) 25 liters

Let x liters of a solution, that is 10% alcohol by volume be added.

10% of x + 50% of 2 = 15% of (x+2)

$$\frac{10}{100} * x + \frac{50}{100} * 2 = \frac{15}{100} * (x+2) = 14 \text{ (A)}$$

Alligation method.

Alligation table. Write the lower percentage at the left side and higher percentage at the right side in the first row. Write the resultant percentage after mixing as mean or average at the middle row. Write the quantity at the last row against each column.

Percentages	10	50
Mean Percentage		15
Quantity	x	2

Alligation formula.

$$\frac{\text{quantity at lower concentration}}{\text{quantity at higher concentration}} = \frac{\text{higher concentration} - \text{mean concentration}}{\text{Mean concentration} - \text{lower concentration}}$$

$$\frac{x}{2} = \frac{50-15}{15-10} = \frac{35}{5} = 7 \quad (x*1) = 7*2 \quad x=14 \text{ (A)}$$

2. Sea water contains 5% salt by weight. How many kilograms of fresh water must be added to 40 kg of sea water for the salt content of the solution be 2%

(a) 30 (b) 40 (c) 60 (d) 80 (e) 100

Solution:

$$\text{Salt content in 40 kg of sea water} = \frac{5}{100} * 40 = 2 \text{ kg}$$

Let x kg of pure water be added

$$\text{Hence } \frac{2}{40+x} * 100 = 2$$

$$x = 60 \text{ kg (B)}$$

Alligation Method. (Only water concentration)

95	98	100
40	x	

$$\frac{40}{x} = \frac{100-98}{98-95} = \frac{2}{3} \quad 2x = 120, \quad x = 60 \text{ (B)}$$

EXERCISE

1. The average weight of a class of 40 students is 30 and the average weight of a class of 20 students is 15. Find the average weight of both the combined classes.

a) 20 b) 24 c) 25 d) 30

2. The ratio of number of girls to number of boys is 1:2. If the average weight of the boys is 30kg and the average weight of both the boys and girls be 25 kg, then the average weight of the girls is?

a) 20 b) 15 c) 25 d) 30

3. 16 liter of kerosene is mixed with 5 liters of petrol. The price of kerosene is Rs.12 per liter and the price of petrol is Rs.33 per liter. The average price of the mixture per liter is:

a) 15 b) 17 c) 19 d) 20

4. Two varieties of soda water with different prices is mixed in the ratio of 2:3. The price of first soda water is RS.10 per liter while the price of second soda water is Rs.15 per liter respectively. The average price of the mixture (per liter is)?

a) 10 b) 13 c) 15 d) 18

5. In what ratio must a grocer mix two varieties of tea worth Rs.60 a kg and Rs.65 a kg so that by selling the mixture at Rs.68.20 a kg he may gain 10%.
a) 3:2 b) 3:4 c) 3:5 d) 4:5
6. Amit travels 30 minutes at the speed of 25 Km/hr. further he travels 20 minutes at the speed of 40km/hr. Find the average speed
a) 31 b) 29 c) 34 d) 33
7. A mixture of rice is sold at Rs.3.00 per kg. This mixture is formed by mixing the rice of Rs. 2.10 and RS.2.52 per Kg. what is the ratio of price of cheaper to the costlier quality in the mixture. If the profit of 25% is being earned.
a) 2:7 b) 2:3 c) 2:5 d) 2:9
8. A merchant has 840kg of rice part of which he sells at 9% profit and the rest at 16% profit he gains 12% profit . On the whole what is the quantity sold at 16% profit?
a) 300 kg b) 360kg c) 380kg d) 400kg
9. An alloy of zinc and Tin contains 35% of Zinc by weight. What quantity of zinc must be added to 400 pounds of the alloy such that there is 60% of zinc by weight in the final mixture?
a) 250 b) 280 c) 220 d) 280
10. Two vessels A and B contains mixtures of milk and water the ratio of milk and water in these two vessels is 5:2 and 8:5 respectively. In what ratio must these mixture be mixed to form a new mixture containing milk and water in 9:4 ratio?
a) 8:2 b) 9:2 c) 7:2 d) 5:2
11. Deepak has some goats and some hens. The total number of heads is 90 and the total number of feet is 248. What is the number of goats?
a) 56 b) 42 c) 58 d) 34
12. An amount of Rs.780 is distributed among 60 students of a class, such that each boy gets Rs.15 and each girl Rs.10. what is the number of boys?
a) 63 b) 32 c) 38 d) 36
13. If 10 kg of salt costing RS.10 per kg and 5 kg of salt costing Rs.4 per kg are mixed. Find the average cost of the mixture in Rs/kg?
a) 5 b) 10 c) 7 d) 8
14. A can contains mixtures of two liquids A and B in the ratio 7:5. When 9 liter of the mixture is drawn off and replaced with liquid B, the ratio of A and B becomes 7:9 what was the initial quantity of liquid A in the can?
a) 10 b) 20 c) 21 d) 25
15. A container has 40 liters of wine from this container 4 liters of wine is taken out and replaced with water this process is repeated two more times. What will be the final quantity of water in the container?
a) 20.18 b) 15.87 c) 10.84 d) 18.97

ASSIGNMENT

16. If the average weight of a class of students is 15 and the average weight of another class of students is 30 , then find the ratio of the students of the first class to the another class of 30 students when the average weight of both the classes is 25.
a) 1:2 b) 2:1 c) 3:4 d) 4:3

17. A seller has two variety of rice worth Rs.80 per kg and Rs.95 per kg.If he gains 20% profit by selling the mixture of two varieties at Rs.84 then the mixing ratio is
a.7:5 b.5:3 c.7:2 d.5:2
18. A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each of the container so as to get 12 liters of milk such that the ratio of water to milk is 3:5?
a) 6 liters b) 1 liter c) 8 liters d) 7 liters
19. Zinc and copper are in the ratio of 5:3 in 200 gm. of the alloy. How much gm. of copper be added to make the ratio of 3:5?
a) $133 \frac{1}{3}$ b) $\frac{1}{200}$ c) 72 d) 66
20. In what proportion sugar at 3.25 per kg is mixed with sugar at Rs.3.80 per kg. So that the mixture be worth Rs.3.50 per kg.
a) 7:5 b) 8:7 c) 6:5 d) 7:8
21. A container contains 40 liters of milk. From this container 4 liters of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?
a) 26.34lit b) 27.36lit c) 28 lit d) 29.16 lit
22. How many kilogram of sugar costing Rs.9 per kg must be mixed with 27kg of sugar costing Rs.7 per kg? So that there may be a gain of 10% by selling the mixture at Rs.9.24/kg?
a) 36 kg b) 42 kg c) 54kg d) 63 kg
23. How much of salt at 42 paise per kg must be mixed with 25 kg of salt at 24 paise per kg. so that a profit of 25% is obtained on selling the mixture at 40 paise per kg.
a) 20 kg b) 25 kg c) 15 kg d) 18kg
24. A gallon contains a mixture of two liquids A and B is the ratio 7:5.When 9 liters of mixture are drawn off the gallon is filled with B, the ratio of A and B becomes 7:9.How many liters of liquid A was contained by the gallon initially?
a) 10 b) 20 c) 21 d) 25
25. The milkman has two types of milk. In the first container the percentage of milk is 80% and in the second container the percentage of milk is 60%. If the mixes 28 liters of milk of the first container to the 32 liters of milk of the second container. Then the percentage of milk in the mixture is?
a) 70.45 b) 69.33 c) 67.53 d) 65